Powder-Excavator: design brainstorming and prior-art notes

vertical-cloud-lab

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Abstract

This document captures a literature-aware brainstorm for the gantry-mounted "powder excavator" sketched in `powder-excavator-sketch.jpg` and recreated as the labelled subpanels under `docs/figures/`. It is structured roughly like the introduction to a *Digital Discovery* manuscript on a new powder-handling tool: where does this device sit in the landscape of automated powder dispensing, what gap is it trying to fill, and what would we need to measure to defend that claim. The current revision adopts the corrected *longitudinal-pivot, sideways-tilt* geometry identified by the Edison Scientific design-review tasks [8], and adds a cam-slot / pin-defined-path actuation variant (Sec. 4) based on a follow-on hand sketch.

Provenance. This brainstorm was prompted by feedback on PR #2 to "send the idea to Edison for a high-effort literature search" and to consider the discussion at the Accelerated Discovery thread [1]. The Edison Scientific service was wired in via `edison_client` for a follow-up session: a high-effort PaperQA3 literature search and two high-effort design-review tasks were run [8], and their verbatim responses are archived under `docs/edison/`. Peer-reviewed citations marked [21, 22, 3, 4, 13, 20, 19, 9, 15, 14, 5, 2, 12, 7, 17, 10, 11, 16] are taken directly from the Edison literature task and are listed in the References section.

Geometry note (post-review). The mechanism described in Sec. 3 and Sec. 6 has been updated to the **longitudinal-pivot, sideways-tilt, smooth-cam-ramp** geometry adopted after the Edison design-review tasks identified two design-blocking issues with the earlier "ferris-wheel gondola" (transverse pivot, end-over-end tilt, sawtooth and hooked-lip) revision: a kinematic impossibility in the push-to-tilt mechanism and a trapped-volume / bridging problem for cohesive powders. Section 4 adds a *pin-defined-path* actuation variant in which a peg on a stem descending from the gantry rides in a routed slot in a fixed external board; this captures the cam-ramp idea in a fully constrained "peg in slot" form that removes the approach/contact ambiguity of a free cam follower.

1 Why powder handling is the pinch point in self-driving labs

Self-driving laboratories (SDLs) for chemistry and materials science have matured rapidly over the past five years, with closed-loop platforms now routinely combining robotic execution, automated characterization, and Bayesian or generative experiment design [21, 20, 9, 16]. Across nearly every published SDL, **solids handling — and powder dispensing in particular — is repeatedly called out as the dominant bottleneck**: it is the step most likely to be left to a human, the one most sensitive to material properties (cohesion, hygroscopicity, triboelectric charging), and the one that most often limits end-to-end autonomy [21, 22, 5, 13]. Tom et al.'s 2024 *Chemical Reviews* survey of SDLs makes the point bluntly: solid-dispensing hardware "remains rare and costly relative to liquid handling, and many labs simply avoid direct powder manipulation" wherever pre-dissolved stocks will do [21].

Compared with liquid handling, powder dispensing is hard for three coupled reasons:

1. **No universal flow regime.** Free-flowing crystalline powders, fluffy nanopowders, sticky organics, and hygroscopic salts each demand different feeders, hoppers, and tip geometries [22, 3, 5].

2. **Dose accuracy scales poorly down.** Sub-milligram gravimetric dosing needs a draft-shielded balance, vibration isolation, and feedback control; the 17,797-dispense ETC benchmark across four commercial platforms found that 2 mg targets exhibited 190 to 680 % higher %RSD than 50 mg targets, and that across the full dataset the percent error spanned −10 % to 3245 % (95.3 % of dispenses fell between −25 % and 100 %) [3, 4].

3. **Cross-contamination is expensive.** Reservoirs and end-effectors must either be disposable, easily cleaned, or dedicated per material, which blows up the part count of a multi-reagent platform [21, 5].

2 Prior art (representative, not exhaustive)

Table 1 summarises the device classes that any new powder-handling tool has to position itself against. The rows are not intended as a complete vendor list; they are the classes of mechanism that recur across the SDL literature [21, 22, 3, 4, 13, 20, 9, 15, 14, 5, 2].

Table 1: Prior-art device classes for automated powder handling. Quantitative claims in the "Strengths" and "Limits" columns are taken from the cited sources.

Class	Strengths	Limits relevant to us
Gravimetric vibratory-head dispensers (Quantos QX1/QX5, Chemspeed SWING / Powdernium / FLEX GDU-Pfd, Unchained Labs Freeslate/Junior)	High precision on dense free-flowing powders; closed-loop weight feedback; integrated workflow software; ETC benchmark established Quantos as the best overall % error / %RSD / time balance and Chemspeed SWING as the fastest (13 to 65 s) [3, 4]	Per-reagent consumable cartridges/heads; abrasive solids damage Quantos heads; hygroscopic powders clog them; Chemspeed SWING and one Unchained system *failed outright* on fumed silica in the ETC benchmark [3, 5]
Auger / screw feeders on robot arms (Berlinguette & MacLeod "Ada" thin-film SDL; Aspuru-Guzik group platforms; Chemspeed FLEX GDU-Pfd hopper/auger module)	Robust to mildly cohesive powders; metered dosing; ±10 % tolerance setting on GDU-Pfd [4]	Mechanical complexity; cleaning between materials non-trivial; the Lunt modular-workflow paper notes "grinding, mixing and recovering solids remains an unsolved challenge in commercial automation" [15, 14]
Spatula-mimicking dual-arm robots (Cooper group's PowderBot [13])	Versatile across material classes; ∼0.07 % error at 200 mg for non-challenging solids; comparable failure rate to commercial systems (23.1 % vs. 26.9 % vs. 26.5 %) at lower hardware cost [13]	Slow throughput due to manipulator motions and corrective restarts; struggles with compressible powders ($CaCO_3$) and highly hygroscopic materials [13]

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Table 1 – continued from previous page

Class	Strengths	Limits relevant to us
Slurry-conversion workflows (Ceder group A-Lab: Quantos doses precursors into vials, ball-milled with ethanol into pumpable slurry) [20, 19, 9]	Sidesteps last-mile dry-transfer issues; well-characterised end-to-end workflow; 36 novel compounds in 17 days	Adds ethanol handling, drying, and a dedicated `RecoverPowder` regrinding task downstream; upstream powder loading is still manual [19]
Capsule / cartridge "drop a pre-weighed slug"	Eliminates real-time dosing error	Needs upstream weighing; Quantos cartridge ecosystem in particular criticised for high cost, single-vendor procurement, RFID-locked reuse, and clogging on hygroscopic powders [5]
Acoustic / ultrasonic non-contact transfer (Echo-style platforms; ultrasonic capillaries, 14 to 16 µg through 0.2 mm nozzles) [22]	Non-contact, no cross-contamination, very high throughput	Limited to specific particle sizes / suspensions; mostly demonstrated in solid-freeforming / additive-manufacturing contexts, not chemistry SDLs
Vacuum / electrostatic pickup (DryPette, Electronic Spatula, automated electrostatic pipettes) [22, 2]	Gentle on the powder; good for fragile crystals; 0.3 mg-class doses possible	Hard to release a clean dose; *exquisitely* sensitive to humidity, charge decay, and powder conductivity [22]
Positive-displacement "PowderPicking" (standard Gilson MICROMAN pipettes with disposable capillary-piston tips, Alsenz) [2, 6]	Cheap; 0.6 to 25 mg range; mean CV ∼6 % across 10 powders; disposable tips eliminate carryover	Manual / medium throughput; depends on complete homogeneous filling of the probe cylinder; tip clogging with adhesive powders [2]
Bulk "scoop and dump" mechanical scoops (**this project**; closest commercial analogue: MTI PF-A glass dispenser referenced in the community thread) [18]	Cheap, mechanically simple, no actuators on the end-effector, easy to clean; the present design has *no consumable* at all	Not a precision dosing tool; dose CV likely 5 to 20 % depending on whether a strike-off bar is used; not for sub-mg work

Two recent overviews are particularly relevant for situating a new device in this landscape: Tom et al.'s 2024 *Chemical Reviews* survey [21] and Lunt et al.'s solid-state-workflow papers [15, 14]. Both flag *bulk powder transfer from stock containers into a precision dispenser* as a step that is still almost universally manual. On the frugal-hardware side, recent open-hardware platforms have demonstrated cost reductions of up to 94 % relative to commercial analogues for adjacent dispensing problems (pellet dispensers, 3D-printed autosamplers) [12, 7], albeit for liquid or pellet handling rather than fine inorganic powders.

2.1 Community prior art from the Accelerated Discovery thread

The Accelerated Discovery thread "Accurate powder dispensing for chemistry and materials science applications" [1] is a useful, opinionated cross-section of what practitioners have actually tried. The points most relevant to our design:

- **Framing (sgbaird, post 1).** Accurate powder dispensing is "around an order of magnitude more complex and more expensive than liquid handling." This matches the SDL surveys [21] and is the gap we are trying to chip away at on the cheap, mechanical end of the design space.
- **3D-printed calibrated spatulas (post 2, after Cook et al. [6]).** Sets of spatulas with known volumes give a calibration curve from volume to mass; decent accuracy in isolation, but degraded by static in gloveboxes and by lighter powders. Two takeaways for us: (i) **a strike-off bar is essentially required** ("just manually tapped the spatula until each scoop is level ... it doesn't seem too difficult to physically level it by passing something against the top of the scoop") — directly endorses the strike-off bar variant in Sec. 6; (ii) per-material calibration is realistic if the trough is geometrically well-defined.
- **Positive-displacement-pipette method (post 3, Alsenz [2]).** Packs a pipette tip into the powder; spans 0.6 to 20 mg with ∼10 % CV across tip sizes. The 10 % CV is a **useful baseline target** for our scoop on cohesive powders.
- **Auger / screw-feed builds (loppe35, post 4).** Cheap (jewelry- scale load cell ≈ USD 25, stepper + control board) but "leaky when dispensing vertically" and needs hopper flow aids. Reinforces keeping the excavator's bucket *un-augered* in the baseline design; the auger is positioned in Sec. 6 as an optional add-on, not the core mechanism.
- **The "pick it back up" problem (muon, posts 5/7/14).** Dispensing is only half the workflow; recovering powder *after* an operation (ball-milling, acoustic mixing, weighing) is at least as hard, and a vacuum/pipette approach is awkward when the milled powder is fine and multi-precursor cross-contamination matters.
- **Autotrickler v4 + A&D FX-120i (sgbaird, post 6).** ±1 mg, fast-response scale, BLE-controllable from a Pico W; community-validated reference for the "scale + metered feed" downstream of a bulk-transfer device. Suggests a concrete integration target: excavator → tared Autotrickler-style trickler/scale.
- **PowderBot (Cooper group) [13] and A-Lab (Ceder group) [20, 9].** Two published end-to-end SDLs that *did* solve solids handling, but with workflow-specific workarounds (typically involving milling).
- **Manual MTI PF-A glass dispenser (sgbaird, post 24 — the specific post linked in the review feedback) [18].** A 250 mL hand dispenser whose geometry is reminiscent of Owen Melville's earlier scoop. The closest existing artifact to our bucket and effectively a manual ancestor of the excavator's scoop motion.

3 Where the powder-excavator fits

The proposed mini excavator is **not** competing with Quantos-class precision dispensers. It is aimed at the **upstream, bulk-transfer step**: moving a relatively coarse aliquot of powder from a stock bed (e.g. a tray, jar, or hopper) to a downstream station — which might itself be a precision dispenser, a sieve, a press, a crucible, a dissolution vial, or simply a weigh boat for a human-in-the-loop workflow. In that niche, the design choices shown in Fig. 1 (post-review) make sense:

- **Half-cylinder ladle suspended on a longitudinal pivot pin.** The pin runs along the trough's long axis L, through the two short end caps, with the two arms gripping those end caps. The trough rolls *sideways* about the pin to pour, dispensing over the *full long edge* rather than through a narrow end-cap point. This eliminates the trapped-volume / arching failure mode that an end-over- end tilt of a half-cylinder would have for cohesive powders (Edison v2 §3, archived in `docs/edison/`).

- **Pure mechanical actuation via a fixed smooth inclined cam ramp.** Removes the servo and its wiring/maintenance burden; the deposit angle is set by how far up the cam the trough's chamfered bumper has ridden (i.e. by gantry X position), not by a torque setpoint. A smooth ramp — rather than a sawtooth tooth — is required because a fixed point engaging a fixed-radius hook on a horizontally-translating pivot is geometrically impossible (Edison v2 §1). Sec. 4 below describes a stricter "peg in routed slot" refinement of this idea.

- **Mandatory bed-edge strike-off bar.** Promoted from "optional variation" to a baseline-required part: without a level-defining strike-off, dose CV will sit in the 15 to 30 % range across powder classes, which would put the device below the $\approx 10\,\%$ CV baseline achievable with Alsenz's positive-displacement-pipette method [2] and below the Cooper PowderBot's 0.07 % at 200 mg [13].

- **Open top with no lid.** Trades some retention loss for radically simpler cleaning — which, per [21, 5], is the single biggest hidden cost in multi-material SDLs.

- **Optional grounded conductive lining.** Because the pivot pin is a metal dowel, it can double as the ground path for an inexpensive copper-tape lining of the trough interior. This is the prototype-friendly equivalent of using ESD-safe filament and dramatically reduces triboelectric charging on fine inorganics (Edison v1 §2).

A reasonable framing for a write-up: *"A purely mechanical, gantry-mounted ladle for bulk powder transfer in self-driving materials labs, designed to bridge stock containers and precision dispensers without adding servo count or cleaning burden, and to take over the upstream-loading and post-process recovery roles flagged as unsolved by Lunt et al."* [15, 14].

The design intentionally makes **no claim** of being a precision dispenser, and tempers the "bidirectional dispense + recover" claim made in earlier framing: a half-cylinder cannot scoop cleanly out of a flat bed without leaving a trail of retained powder, so recovery is a meaningful but lossy operation rather than a complete one (Edison v2 §7).

4 Pin-defined-path actuation: a routed-slot refinement

A hand-sketch (`PXL_20260423_231729467.jpg`) suggests an actuation refinement that fully resolves the "approach-and-contact" ambiguity of the smooth cam ramp described in Sec. 3. Rather than a free cam follower that approaches a fixed inclined surface and slides up it, the trough is suspended from the gantry carriage by a short *stem* that carries a transverse *peg* at its top end. This peg rides in a *routed slot* cut into a fixed external board mounted at fixed height in the work area (Fig. 2). The carriage–stem joint is a pin pivot; the peg is captured in the slot for the entire stroke. Pure-X gantry travel therefore drives a deterministic tilt-vs.-X schedule given by the slot's routed path: where the slot is straight and horizontal the trough holds level (scoop, transport); where it deviates, the difference between the carriage's straight-line path and the slot's routed path forces the stem (and the trough rigidly attached to it) to rotate about the carriage pivot.

This is mechanically the same idea as the cam paths used in coin-op mechanisms, player pianos, and Geneva-style indexing devices, applied to the gantry geometry of a self-driving lab.

Compared with the smooth inclined cam ramp of Sec. 3, the pin-in-slot refinement has several concrete advantages:

- **No engagement event.** The peg never has to "find" the cam; it is captive in the slot from the moment the trough is mounted. There is no lift-off direction in which the contact can lose registration.
- **Bidirectional and reversible.** The slot path constrains motion in both $+X$ and $-X$ directions, so the trough can be tipped, held, and *returned to level* purely by gantry motion. The smooth cam ramp relies on gravity for the return.
- **Programmable schedule.** The slot is the program. Different routed slots — straight-then-rising, multi-stage with a hold detent, a small "knock" wiggle to dislodge bridges (Edison v1 §6) — implement different dispensing schedules without any change to the trough or carriage. The strike-off bar sweep can be encoded as a small detent in the slot near the bed-edge X coordinate.
- **Deterministic tilt at deposit.** Because the peg is captured on both sides of the slot wall, the deposit angle is set by slot geometry to the tolerance of the routing operation, not by friction at a sliding cam interface.
- **Simpler statics analysis.** The slot wall provides both the tilt-up reaction force and (on the return stroke) the tilt-down force. The smooth cam ramp only provides the former; the latter relies on gravity, which fails for very light troughs or when the trough is empty.

Caveats. The slot must be cut to the same accuracy as the finished tilt schedule requires; a cleanly milled slot in acrylic or aluminium is straightforward, but FDM-printed slots will have layer-line ripples that the peg will feel as a low-amplitude tilt jitter. The peg should be a hardened dowel pin with a low-friction sleeve (PTFE or oil- impregnated bronze) to minimise stick-slip, and the slot should be slightly oversized in the direction perpendicular to the desired motion to allow thermal play without binding. The board itself becomes a second *registered* fixture in the work area — it must be mounted in a known relationship to the gantry's X axis, which is the same registration problem the bed and the deposit station already face, and so does not introduce a new class of failure.

5 Open questions and what we'd actually need to measure

To defend the design in a manuscript-style argument we would need at minimum:

- **Dose statistics across powder classes.** Mean and CV of delivered mass for at least three archetypal powders (e.g. free-flowing crystalline like NaCl, cohesive metal-oxide nanopowder like TiO_2, hygroscopic salt like LiCl) across at least 30 scoops each. Following Cooper-group practice [13], this should also include the known "hard" cases — fumed silica (which broke the Chemspeed SWING and one Unchained system in the ETC benchmark [3]) and $CaCO_3$ (which broke the PowderBot [13]).
- **Standardised flowability descriptors per powder.** Report Carr's compressibility index (CI $\leq 10\,\%$ = excellent; 21 to 25 % = passable; >35 % = very poor) and the Hausner ratio (HR 1.00 to 1.11 = excellent; 1.26 to 1.34 = passable; >1.45 = poor) for every test powder, plus Freeman FT4 basic flow energy (BFE) and specific energy (SE) where available [17, 10, 11].
- **Carryover / cross-contamination.** Mass remaining in the trough after a "tip and pour" event vs. after a programmed cleaning cycle (knock-oscillation against the cam, inverted shake, brush, or air blast), stratified by powder class, following the approach Carruthers used to expose Quantos-cartridge limitations [5].

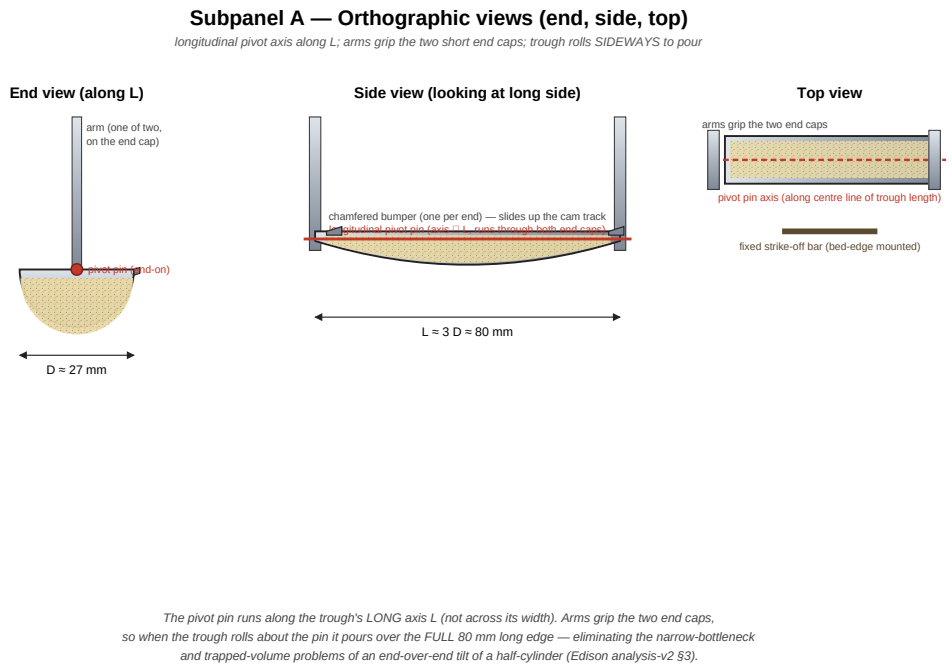

Figure 1: Orthographic views of the half-cylinder trough with the longitudinal pivot pin, end-cap pivot bosses, and the chamfered bumper that engages the cam ramp. Generated by `scripts/generate_figures.py`.

- **Sensitivity to bed depletion.** How much does the dose change as the powder bed depletes / gets cratered? Run 30 consecutive scoops from the same X, Y location and plot dispensed mass vs. scoop number (Edison v1 §3). The expected steep decay curve will set the requirement that the gantry must raster across the bed.
- **Coupling to a downstream balance.** Demonstrate that the excavator → balance → vibratory-head pipeline reduces total wall-clock time per dispensed dose vs. a one-stage precision dispenser doing both bulk and fine work — i.e. directly attack the upstream-loading bottleneck flagged by [21, 15, 14, 5].
- **Mandatory benchmark experiments (Edison v2 §6).**
  1. **Cam-engagement test.** 3D print the trough, the chamfered bumper, and the inclined cam track. Drive the gantry purely in X and verify the bumper rides up the cam without binding, skipping, or requiring coordinated Z motion.
  2. **Pin-slot equivalence test.** Repeat the same dispense schedule with the cam ramp of Sec. 3 and the routed slot of Sec. 4; compare tilt-angle repeatability and dose CV.
  3. **Trapped-volume bake-off.** Compare retained mass of a cohesive test powder (TiO_2 or flour) for a sideways tilt vs. a hypothetical end-over-end tilt of the same trough. The expected 7 to 26 % retention penalty of the end-over-end case (Edison v2 §3) is the empirical justification for the longitudinal-pivot geometry.
  4. **Strike-off efficacy.** Measure the dose mass of 30 consecutive scoops with and without the bed-edge strike-off bar. The expected 2–3× CV improvement is the empirical justification for keeping the strike-off bar in the baseline.
- **Cam-ramp / strike-off / bumper geometry sweep.** Cam slope, ramp length, strike-off cross-section, and rim-bumper chamfer angle are the four knobs most likely to dominate dose CV; small parameter sweeps are cheap with 3D-printed parts. For the pin-slot variant, the equivalent sweep is over slot path shape, peg diameter, and peg sleeve material.

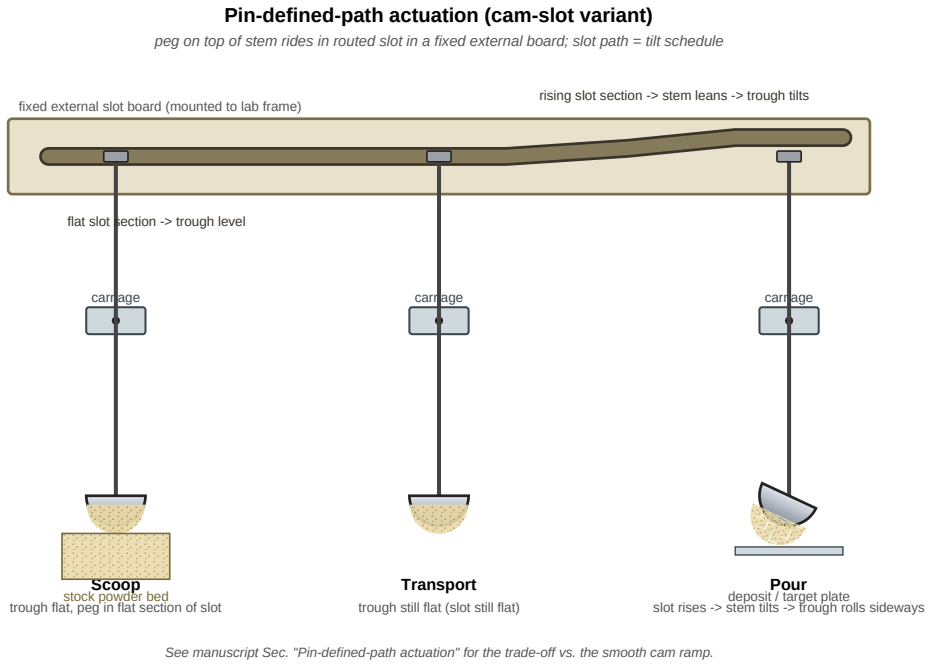

Figure 2: Pin-defined-path actuation variant (Sec. 4). A vertical stem hangs from the gantry carriage on a pivot; a transverse peg at the top of the stem rides in a routed slot cut into a fixed external board. The slot's shape over X is the program for the trough's tilt schedule. Generated by `scripts/generate_figures.py`.

6 Design variations worth prototyping (still pure-mechanical)

Carrying over from the original brainstorming, with the literature in mind:

- **Pin-defined-path actuation (Sec. 4).** Replaces or augments the smooth cam ramp; the slot board is the new fixed-fixture part. Worth prototyping side-by-side with the cam ramp to compare schedule programmability and dose CV.
- **Dual cam ramps (or dual slot boards), left and right** — for two-output workflows (e.g. one for "real" deposit, one for "purge / waste" to clear residual material between materials).
- **Strike-off bar variants** — a square cross-section vs. a chamfered blade vs. a thin compliant flap, scored for fill repeatability and for how cleanly the bar sweeps the rim of cohesive powders without flicking material away.
- **Twin troughs back-to-back** — fill one while dumping the other for doubled throughput at the cost of one extra carriage stop.
- **Replaceable, snap-in trough liners** — a thin, single-material liner (e.g. ESD-safe filament or copper-tape-lined PETG) that snaps into the printed trough body. Lets us treat the trough as consumable per-material and bypass the cleaning problem entirely; this is the prototype-friendly analogue of Quantos cartridges [5] without the RFID lock-in.
- **Spring-loaded passive flap lid actuated by the same cam or slot** — closes during transport (curing the fine-powder retention problem) and opens at the same instant the cam (or slot detent) engages, with no extra actuator.

- **Internal auger as an *option*** — adds one rotary actuator but converts the device from "tilt and pour all" to a coarse metered dispenser, which would squarely overlap with mid-range commercial systems [4].

References

[1] Accelerated Discovery community. Accurate powder dispensing for chemistry and materials science applications. Online discussion thread, 2024. URL: https://accelerated-discovery.org/t/accurate-powder-dispensing-for-chemistry-and-materials-science-applications/177.

[2] Jochem Alsenz. PowderPicking: an inexpensive, manual, medium-throughput method for powder dispensing. *Powder Technology*, 209(1–3):152–157, May 2011. doi:10.1016/j.powtec.2011.02.014.

[3] Matthew N. Bahr, David B. Damon, Simon D. Yates, Alexander S. Chin, J. David Christopher, Samuel Cromer, Nicholas Perrotto, Jorge Quiroz, and Victor Rosso. Collaborative evaluation of commercially available automated powder dispensing platforms for high-throughput experimentation in pharmaceutical applications. *Organic Process Research & Development*, 22(11):1500–1508, October 2018. doi:10.1021/acs.oprd.8b00259.

[4] Matthew N. Bahr, Mark A. Morris, Noah P. Tu, and Aakankschit Nandkeolyar. Recent advances in high-throughput automated powder dispensing platforms for pharmaceutical applications. *Organic Process Research & Development*, 24(11):2752–2761, October 2020. doi:10.1021/acs.oprd.0c00411.

[5] Stephen Carruthers. *A Mobile Robotic Researcher for Autonomous Solar Fuels Research*. PhD thesis, University of Liverpool, 2025. Cited via Edison Scientific PaperQA3 (see `docs/edison/`); pp. 115–119 document practical Quantos cartridge limitations (cost, vendor lock-in, RFID-limited reuse, hygroscopic clogging, fragile internals).

[6] C. J. Cook, A. Q. Fenwick, K. Galvin, A. Mizes, S. E. Reisman, B. M. Stoltz, J. T. Behaghel de Bueren, M. Kammler, B. Hu, T. D. P. Stack, G. H. Mehl, A. M. Quintana, A. Salinas, J. F. Pinón-Hernández, K. N. Houk, E. M. Carreira, A. B. Smith, and S. T. Phillips. A guide to the inert-atmosphere processing of sol–gel reactions. *Nature Protocols*, 16:365–388, 2021. doi:10.1038/s41596-020-00452-7.

[7] Sayan Doloi, Maloy Das, Yujia Li, Zen Han Cho, Xingchi Xiao, John V. Hanna, Matthew Osvaldo, and Leonard Ng Wei Tat. Democratizing self-driving labs: advances in low-cost 3D printing for laboratory automation. *Digital Discovery*, 4:1685–1721, January 2025. doi:10.1039/D4DD00411F.

[8] Edison Scientific. Edison Scientific AI platform: literature search (PaperQA3) and design-review tasks. AI service responses; verbatim raw answers archived in `docs/edison/`, 2026. URL: https://edisonscientific.com/.

[9] Yuxing Fei, Bernardus Rendy, Rishi Kumar, Olympia Dartsi, Hrushikesh P. Sahasrabuddhe, Matthew J. McDermott, Zheren Wang, Nathan J. Szymanski, Lauren N. Walters, David Milsted, Yan Zeng, Anubhav Jain, and Gerbrand Ceder. AlabOS: A Python-based reconfigurable workflow management framework for autonomous laboratories. *arXiv preprint arXiv:2405.13930*, May 2024. doi:10.48550/arXiv.2405.13930.

[10] Tim Freeman and Katrina Brockbank. Characterising powder flow properties – the need for a multivariate approach. *EPJ Web of Conferences (Powders and Grains)*, 2017. Cited

via Edison Scientific PaperQA3 (see `docs/edison/`). URL: https://www.freemantech.co.uk/about-powder-flow.

[11] Chinmay Ghoroi, Lakxmi Gurumurthy, D. J. McDaniel, Laila J. Jallo, and Rajesh N. Davé. Multi-faceted characterization of pharmaceutical powders to discern the influence of surface modification. *Powder Technology*, 236:63–74, February 2013. doi:10.1016/j.powtec.2012.05.039.

[12] Miguel Hernández-del Valle, Jorge Ilarraza-Zuazo, Enrique Dios-Lázaro, Javier Rubio, Joris Audoux, and Maciej Haranczyk. Pellet dispensomixer and pellet distributor: open hardware for nanocomposite space exploration via automated material compounding. *Digital Discovery*, July 2024. doi:10.1039/D4DD00198B.

[13] Ying Jiang, Hatem Fakhruldeen, Gabriella Pizzuto, Louis Longley, Ai He, Tianwei Dai, Rob Clowes, Nicola Rankin, and Andrew I. Cooper. Autonomous biomimetic solid dispensing using a dual-arm robotic manipulator. *Digital Discovery*, 2(6):1733–1744, 2023. doi:10.1039/D3DD00075C.

[14] Amy M. Lunt. *A robotic workflow to screen crystal forms using powder X-ray diffraction*. PhD thesis, University of Liverpool, 2024. doi:10.17638/03178518.

[15] Amy M. Lunt, Hatem Fakhruldeen, Gabriella Pizzuto, Louis Longley, Alexander White, Nicola Rankin, Rob Clowes, Ben Alston, Lucia Gigli, Graeme M. Day, Andrew I. Cooper, and Sam Y. Chong. Modular, multi-robot integration of laboratories: an autonomous solid-state workflow for powder x-ray diffraction. *arXiv preprint arXiv:2309.00544*, January 2023. doi:10.48550/arXiv.2309.00544.

[16] Eric McCalla. Semiautomated experiments to accelerate the design of advanced battery materials: Combining speed, low cost, and adaptability. *ACS Engineering Au*, 3(6):391–402, November 2023. doi:10.1021/acsengineeringau.3c00037.

[17] K. K. Moravkar, S. D. Korde, B. A. Bhairav, S. B. Shinde, S. V. Kakulade, and S. S. Chalikwar. Traditional and advanced flow characterization techniques: a platform review for development of solid dosage form. *Indian Journal of Pharmaceutical Sciences*, January 2020. doi:10.36468/pharmaceutical-sciences.726.

[18] MTI Corporation. Manual dispenser of 250 ml made of glass for solid powder, model PF-A. Vendor product page and demonstration video, 2024. Referenced in the Accelerated Discovery community thread (post 24) as a manual analogue of the gantry-mounted scoop. URL: https://www.youtube.com/watch?v=Ur7GZV2PaPQ.

[19] Nathan J. Szymanski. *Automating the Synthesis and Characterization of Inorganic Materials*. PhD thesis, University of California, Berkeley, 2024. Cited via Edison Scientific PaperQA3 (see `docs/edison/`); pp. 78–80 describe the A-Lab manual stock-bottle loading step.

[20] Nathan J. Szymanski, Bernardus Rendy, Yuxing Fei, Rishi E. Kumar, Tanjin He, David Milsted, Matthew J. McDermott, Max Gallant, Ekin Dogus Cubuk, Amil Merchant, Haegyeom Kim, Anubhav Jain, Christopher J. Bartel, Kristin Persson, Yan Zeng, and Gerbrand Ceder. An autonomous laboratory for the accelerated synthesis of novel materials. *Nature*, 624:86–91, November 2023. doi:10.1038/s41586-023-06734-w.

[21] Gary Tom, Stefan P. Schmid, Sterling G. Baird, Yang Cao, Kourosh Darvish, Han Hao, Stanley Lo, Sergio Pablo-García, Ella M. Rajaonson, Marta Skreta, Naruki Yoshikawa, Samantha Corapi, Gun Deniz Akkoc, Felix Strieth-Kalthoff, Martin Seifrid, and Alán Aspuru-Guzik. Self-driving laboratories for chemistry and materials science. *Chemical Reviews*, 124(16):9633–9732, August 2024. doi:10.1021/acs.chemrev.4c00055.

[22] Shoufeng Yang and Jörg Evans. Metering and dispensing of powder; the quest for new solid freeforming techniques. *Powder Technology*, 178(1):56–72, September 2007. doi:10.1016/j.powtec.2007.04.004.